

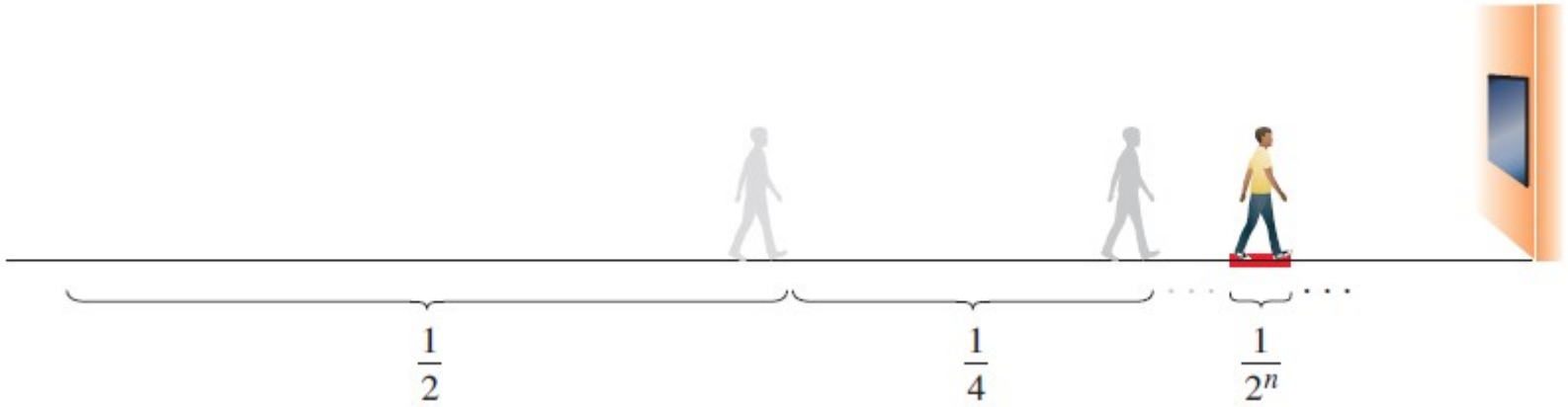
Chapter 9:

Infinite Series

Section 9.1: Sequences

Zeno's Paradox (Dichotomy Paradox)

$$\frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{8}, \quad \frac{1}{16}, \quad \frac{1}{32}, \quad \cdots, \quad \frac{1}{2^n}, \quad \cdots$$



Q) Can we take the sum of infinite numbers?

Infinite Sequences

An **infinite sequence**, or just a **sequence**, can be thought of as a list of numbers written in a definite order:

$$a_1,\ a_2,\ a_3,\ a_4,\ \ldots,\ a_n,\ \ldots$$

The number a_1 is called the *first term*, a_2 is the *second term*, and in general a_n is the *nth term*. We will deal exclusively with infinite sequences and so each term a_n will have a successor a_{n+1}.

Notice that for every positive integer n there is a corresponding number a_n and so a sequence can be defined as a function f whose domain is the set of positive integers. But we usually write a_n instead of the function notation $f(n)$ for the value of the function at the number n.

NOTATION The sequence $\{a_1, a_2, a_3, \ldots\}$ is also denoted by

$$\{a_n\} \qquad \text{or} \qquad \{a_n\}_{n=1}^{\infty}$$

Unless otherwise stated, we assume that n starts at 1.

EXAMPLE

EXAMPLE 1 Some sequences can be defined by giving a formula for the nth term.

(a) At the beginning of the section we described a sequence of distances walked by a man in a room. The following are three equivalent descriptions of this sequence:

$$\left\{\frac{1}{2^n}\right\} \qquad a_n = \frac{1}{2^n} \qquad \left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \ldots, \frac{1}{2^n}, \ldots\right\}$$

In the third description we have written out the first few terms of the sequence: $a_1 = 1/2^1$, $a_2 = 1/2^2$, and so on.

EXAMPLE

Consider the sequence {–°–º–º_°- - ºs }ʳ Then we can denote it as ɔ ɣ Ⱪ –ə or {–ə}ʳ

EXAMPLE

Consider the sequence $\{1, 3, 5, 7, 9, \ldots\}$. Then we can denote it as $a_n = 2n - 1$ or $\{2n - 1\}$.

If we want to denote the sequence with a different starting number say 0, then we write $\{2n + 1\}_0^\infty$.

EXAMPLE

The definition $\left\{\dfrac{n}{n+1}\right\}_{n=2}^{\infty}$ indicates that the formula for the nth term is $a_n = \dfrac{n}{n+1}$ and we start the sequence with $n = 2$: $\left\{\dfrac{2}{3}, \dfrac{3}{4}, \dfrac{4}{5}, \dfrac{5}{6}, \ldots\right\}$

EXAMPLE

The sequence $\{-1, 1, -1, 1, -1, \dots\}$ can be represented as $a_n = (-1)^n$ or $\{(-1)^n\}$.

If we have the sequence $\{1, -1, 1, -1, 1, \dots\}$ then we write $a_n = (-1)^{n+1}$ or $\{(-1)^{n+1}\}$.

EXAMPLE

The definition $\left\{(-1)^n \frac{(n+1)}{3^n}\right\}_{n=0}^{\infty}$ generates the sequence

$$\left\{\frac{1}{1}, -\frac{2}{3}, \frac{3}{9}, -\frac{4}{27}, \frac{5}{81}, \dots\right\}$$

There are sequences that is impossible to find an explicit formula for them.

EXAMPLE

(a) The sequence $\{p_n\}$, where p_n is the population of the world as of January 1 in the year n.

(b) If we let a_n be the digit in the nth decimal place of the number e, then $\{a_n\}$ is a sequence whose first few terms are

$$\{7, 1, 8, 2, 8, 1, 8, 2, 8, 4, 5, \ldots\}$$

(c) **The Fibonacci sequence** $\{f_n\}$ is defined *recursively* by the conditions

$$f_1 = 1 \qquad f_2 = 1 \qquad f_n = f_{n-1} + f_{n-2} \qquad n \geqslant 3$$

Each term is the sum of the two preceding terms. The first few terms are

$$\{1, 1, 2, 3, 5, 8, 13, 21, \ldots\}$$

This sequence arose when the 13th-century Italian mathematician known as Fibonacci solved a problem concerning the breeding of rabbits (see Exercise 81). ■

EXAMPLE

Finding the *n*th Term of a Sequence **In Exercises 45–52, write an expression for the *n*th term of the sequence and then determine whether the sequence you have chosen converges or diverges. (There is more than one correct answer.)**

48. $1, -\frac{1}{4}, \frac{1}{9}, -\frac{1}{16}, \ldots$

SOLUTION

EXAMPLE

Finding the *n*th Term of a Sequence **In Exercises 45–52, write an expression for the *n*th term of the sequence and then determine whether the sequence you have chosen converges or diverges. (There is more than one correct answer.)**

45. 2, 8, 14, 20, . . .

SOLUTION

The Limit of a Sequence

A sequence can be pictured either by plotting its terms on a number line or by plotting its graph. Figures 2 and 3 illustrate these representations for the sequence

$$\left\{\frac{n}{n+1}\right\} = \left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots\right\}$$

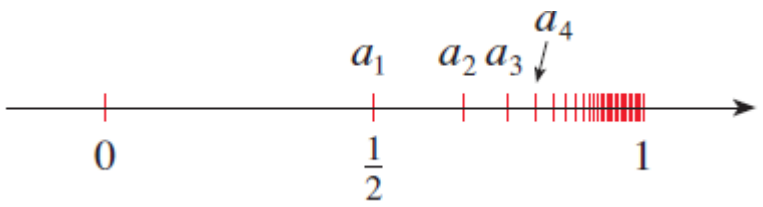


FIGURE 2

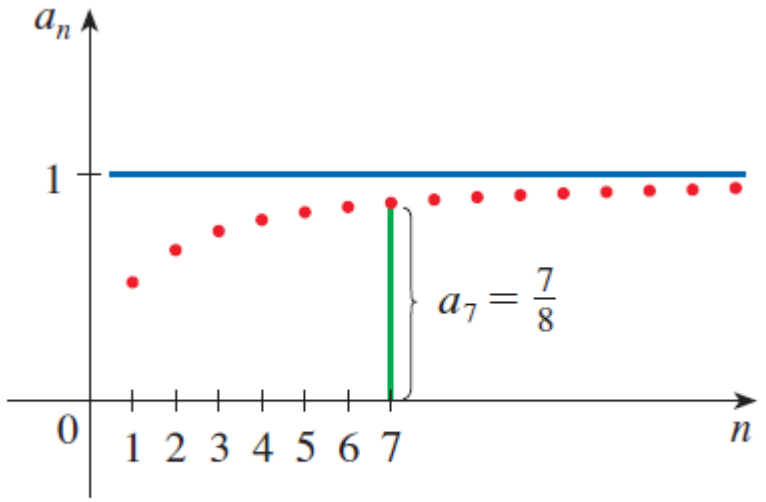


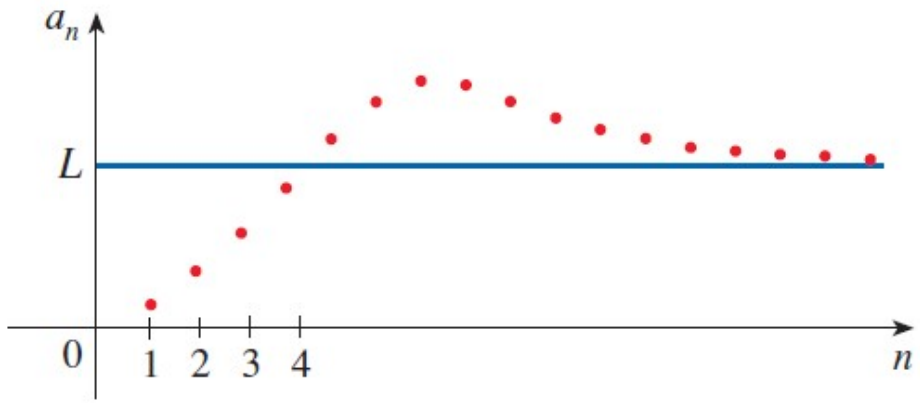
FIGURE 3

The Limit of a Sequence

1 Intuitive Definition of a Limit of a Sequence A sequence $\{a_n\}$ has the **limit** L and we write

$$\lim_{n\to\infty} a_n = L \qquad \text{or} \qquad a_n \to L \text{ as } n \to \infty$$

if we can make the terms a_n as close to L as we like by taking n sufficiently large. If $\lim_{n\to\infty} a_n$ exists, we say the sequence **converges** (or is **convergent**). Otherwise, we say the sequence **diverges** (or is **divergent**).



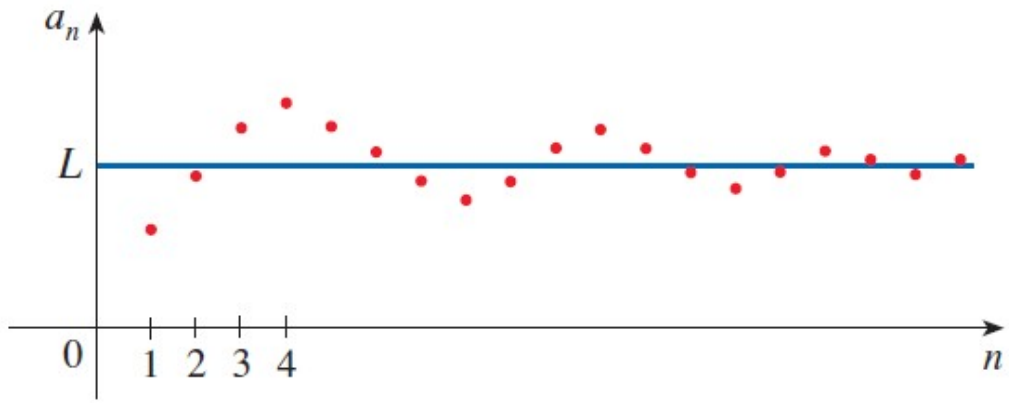


FIGURE 4
Graphs of two convergent sequences with $\lim_{n\to\infty} a_n = L$

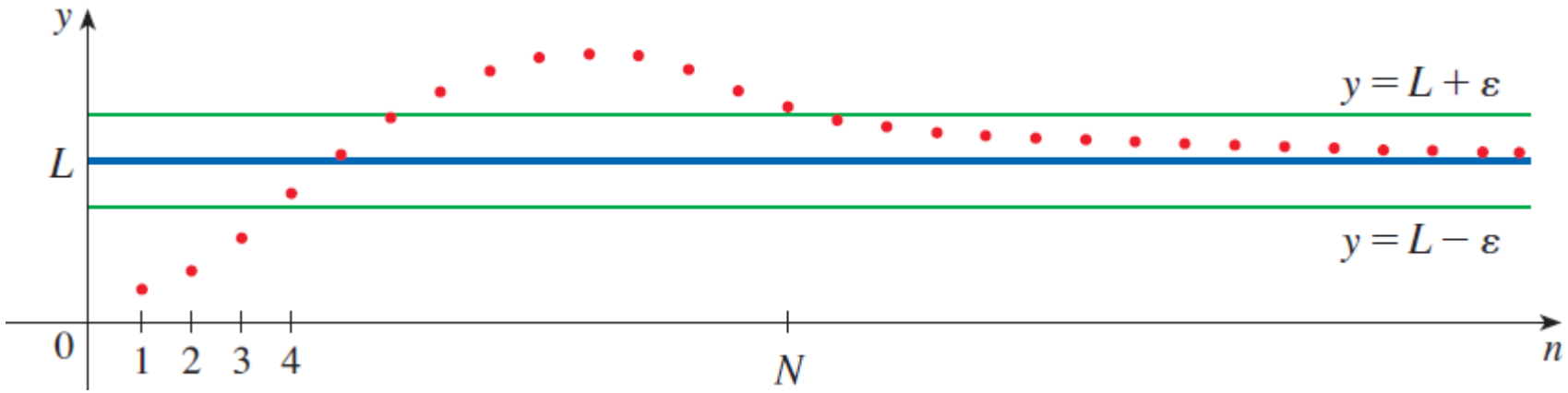
The Limit of a Sequence

2 Precise Definition of a Limit of a Sequence A sequence $\{a_n\}$ has the **limit** L and we write

$$\lim_{n\to\infty} a_n = L \qquad \text{or} \qquad a_n \to L \text{ as } n \to \infty$$

if for every $\varepsilon > 0$ there is a corresponding integer N such that

$$\text{if} \quad n > N \quad \text{then} \quad |a_n - L| < \varepsilon$$



The Limit of a Sequence

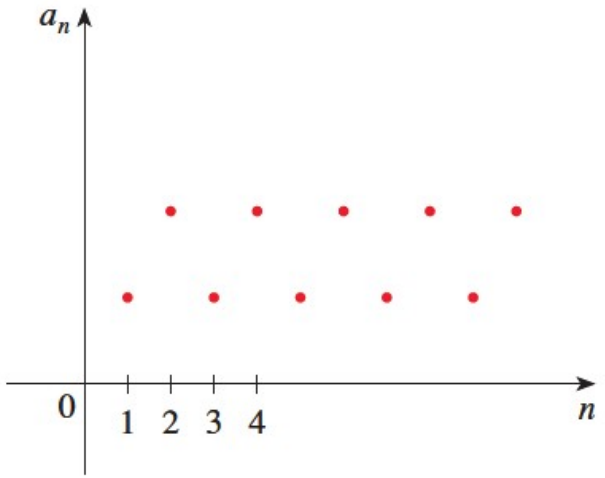
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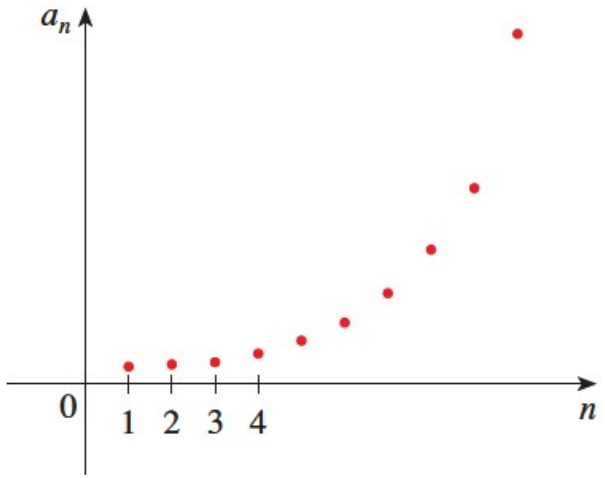
if for every $\varepsilon > 0$ there is a corresponding integer N such that

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Examples of divergent sequences



(a)



(b)

The Limit of a Sequence

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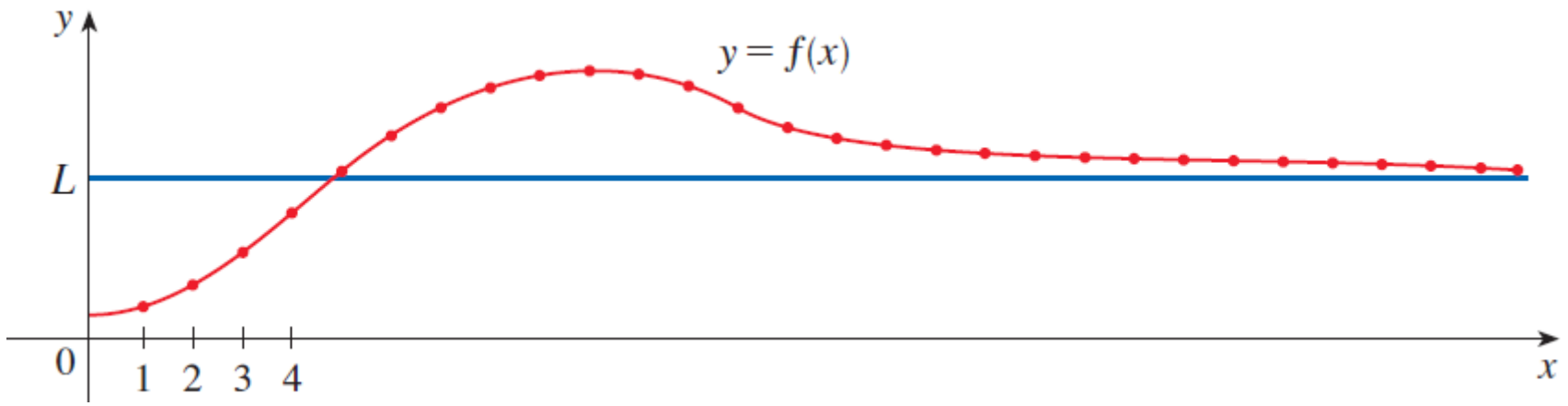
3 Precise Definition of an Infinite Limit The notation $\lim_{n\to\infty} a_n = \infty$ meansthat for every positive number M there is an integer N such that

$$\text{if} \quad n > N \quad \text{then} \quad a_n > M$$

An analogous definition applies for $\lim_{n\to\infty} a_n = -\infty$.

The Limit of a Sequence

4 Theorem If $\lim_{x\to\infty} f(x) = L$ and $f(n) = a_n$ when n is an integer, then $\lim_{n\to\infty} a_n = L$.



For instance, since we know that $\lim_{x\to\infty} (1/x^r) = 0$ when $r > 0$ (Theorem 2.6.5), it follows from Theorem 4 that

5
$$\lim_{n\to\infty} \frac{1}{n^r} = 0 \qquad \text{if } r > 0$$

The Limit Laws given in Section 2.3 also hold for the limits of sequences and their proofs are similar.

The Limit of a Sequence

Limit Laws for Sequences Suppose that $\{a_n\}$ and $\{b_n\}$ are convergent sequences and c is a constant. Then

Sum Law

$$\textbf{1.}\ \lim_{n\to\infty}(a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n$$

Difference Law

$$\textbf{2.}\ \lim_{n\to\infty}(a_n - b_n) = \lim_{n\to\infty} a_n - \lim_{n\to\infty} b_n$$

Constant Multiple Law

$$\textbf{3.}\ \lim_{n\to\infty} ca_n = c \lim_{n\to\infty} a_n$$

Product Law

$$\textbf{4.}\ \lim_{n\to\infty}(a_n b_n) = \lim_{n\to\infty} a_n \cdot \lim_{n\to\infty} b_n$$

Quotient Law

$$\textbf{5.}\ \lim_{n\to\infty} \frac{a_n}{b_n} = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n} \quad \text{if } \lim_{n\to\infty} b_n \neq 0$$

The Limit of a Sequence

Power Law

$$\lim_{n\to\infty} a_n^p = \left[\lim_{n\to\infty} a_n\right]^p \quad \text{if } p > 0 \text{ and } a_n > 0$$

Squeeze Theorem for Sequences

If $a_n \leqslant b_n \leqslant c_n$ for $n \geqslant n_0$ and $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then $\lim_{n\to\infty} b_n = L$.

6 Theorem

If $\lim_{n\to\infty} |a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$.

EXAMPLE

Find $\lim_{n\to\infty} \frac{1}{n}$.

SOLUTION

EXAMPLE

Find $\lim_{n \to \infty} \frac{n}{n+1}$.

SOLUTION

EXAMPLE

Is the sequence $a_n = \dfrac{n}{\sqrt{10 + n}}$ convergent or divergent?

SOLUTION

EXAMPLE

Calculate $\lim_{n \to \infty} \frac{\ln n}{n}$.

SOLUTION

EXAMPLE

Determine whether the sequence $a_n = (-1)^n$ is convergent or divergent.

SOLUTION If we write out the terms of the sequence, we obtain

$$\{-1, 1, -1, 1, -1, 1, -1, \ldots\}$$

The graph of this sequence is shown in Figure 10. Since the terms oscillate between 1 and -1 infinitely often, a_n does not approach any number. Thus $\lim_{n\to\infty} (-1)^n$ does not exist; that is, the sequence $\{(-1)^n\}$ is divergent. ■

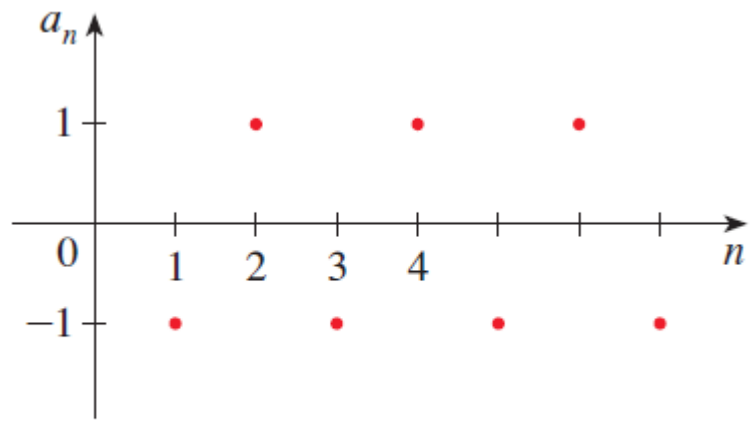


FIGURE 10 The sequence $\{(-1)^n\}$

EXAMPLE

Find $\lim_{n\to\infty}\left((-1)^n\frac{n}{n+1}\right)$ if it exists.

SOLUTION

EXAMPLE

Evaluate $\lim_{n\to\infty} \dfrac{(-1)^n}{n}$ if it exists.

SOLUTION We first calculate the limit of the absolute value:

$$\lim_{n\to\infty} \left| \frac{(-1)^n}{n} \right| = \lim_{n\to\infty} \frac{1}{n} = 0$$

Therefore, by Theorem 6,

$$\lim_{n\to\infty} \frac{(-1)^n}{n} = 0$$

The sequence is graphed in Figure 11.

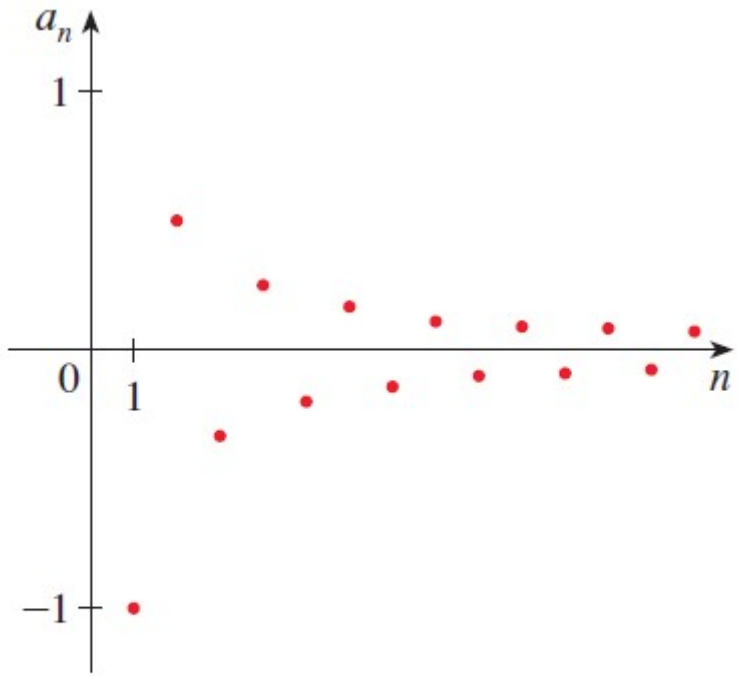


FIGURE 11 The sequence $\left\{ \dfrac{(-1)^n}{n} \right\}$

EXAMPLE Think about it

Find $\lim_{n\to\infty} \frac{\sin n}{n}$.

SOLUTION

Conclusion

$$\lim_{n\to\infty}\left((-1)^n\, a_n\right)\begin{cases} = 0 & \text{if } \lim_{n\to\infty} a_n = 0 \\ DNE & \text{if } \lim_{n\to\infty} a_n = c \neq 0 \text{ or } +\infty \text{ or } -\infty \end{cases}$$

The Limit of a Sequence

7 Theorem If $\lim_{n\to\infty} a_n = L$ and the function f is continuous at L, then

$$\lim_{n\to\infty} f(a_n) = f(L)$$

EXAMPLE

EXAMPLE 9 Find $\lim_{n\to\infty} \sin\frac{\pi}{n}$.

SOLUTION Because the sine function is continuous at 0, Theorem 7 enables us to write

$$\lim_{n\to\infty} \sin\frac{\pi}{n} = \sin\left(\lim_{n\to\infty} \frac{\pi}{n}\right) = \sin 0 = 0$$

■

EXAMPLE

Find $\lim_{n\to\infty} \tan\left(\frac{2n\pi}{1+8n}\right)$.

SOLUTION

The Limit of a Sequence

9 The sequence $\{r^n\}$ is convergent if $-1 < r \leq 1$ and divergent for all other values of r.

$$\lim_{n\to\infty} r^n = \begin{cases} 0 & \text{if } -1 < r < 1 \\ 1 & \text{if } r = 1 \end{cases}$$

EXAMPLE

(a) The sequence $a_n = \left(-\frac{1}{2}\right)^n$ converges to zero since $r = -\frac{1}{2} < 1$. In other words: $\lim_{n\to\infty} \left(-\frac{1}{2}\right)^n = 0$

(b) The sequence $a_n = e^n$ diverges since $r = e > 1$.

❑ Factorial

The symbol $n!$ (read "n factorial") is used to simplify some of the formulas developed in this chapter. Let n be a positive integer; then ***n* factorial** is defined as

$$n! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot \cdots \cdot (n-1) \cdot n.$$

As a special case, **zero factorial** is defined as $0! = 1$. From this definition, you can see that $1! = 1$, $2! = 1 \cdot 2 = 2$, $3! = 1 \cdot 2 \cdot 3 = 6$, and so on. Factorials follow the same conventions for order of operations as exponents. That is, just as $2x^3$ and $(2x)^3$ imply different orders of operations, $2n!$ and $(2n)!$ imply the orders

$$2n! = 2(n!) = 2(1 \cdot 2 \cdot 3 \cdot 4 \cdot \cdots \cdot n)$$

and

$$(2n)! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot \cdots \cdot n \cdot (n+1) \cdot \cdots \cdot 2n$$

respectively.

EXAMPLE

Simplifying Factorials **In Exercises 17–20, simplify the ratio of factorials.**

17. $\dfrac{(n+1)!}{(n-1)!}$

SOLUTION

EXAMPLE

Show that the sequence $\{c_n\} = \left\{(-1)^n \frac{1}{n!}\right\}$ converges, and find its limit.

Solution To apply the Squeeze Theorem, you must find two convergent sequences that can be related to $\{c_n\}$. Two possibilities are $a_n = -1/2^n$ and $b_n = 1/2^n$, both of which converge to 0. By comparing the term $n!$ with 2^n, you can see that

$$n! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot \cdots \cdot n = 24 \cdot \underbrace{5 \cdot 6 \cdot \cdots \cdot n}_{n-4 \text{ factors}} \qquad (n \geq 4)$$

and

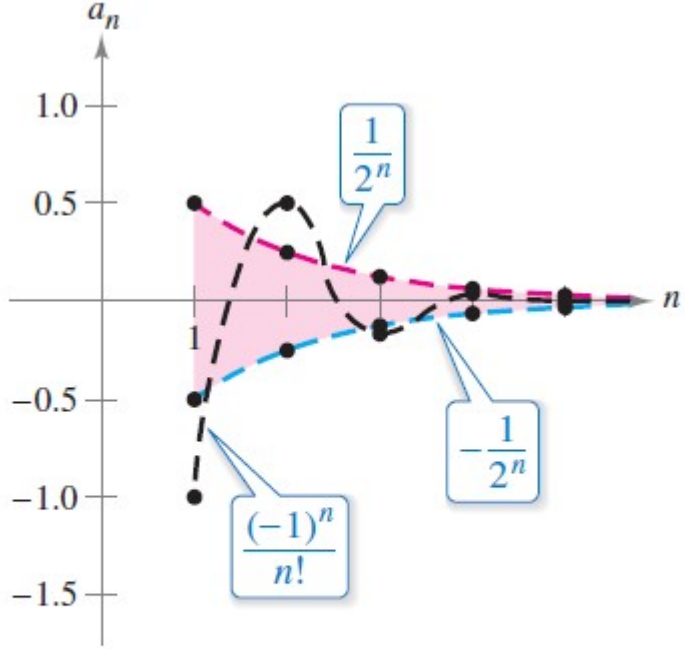
$$2^n = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot \cdots \cdot 2 = 16 \cdot \underbrace{2 \cdot 2 \cdot \cdots \cdot 2}_{n-4 \text{ factors}}. \qquad (n \geq 4)$$

This implies that for $n \geq 4$, $2^n < n!$, and you have

$$\frac{-1}{2^n} \leq (-1)^n \frac{1}{n!} \leq \frac{1}{2^n}, \quad n \geq 4$$

as shown in Figure 9.2. So, by the Squeeze Theorem, it follows that

$$\lim_{n \to \infty} (-1)^n \frac{1}{n!} = 0.$$



For $n \geq 4$, $(-1)^n/n!$ is squeezed between $-1/2^n$ and $1/2^n$.

Figure 9.2

Monotonic and Bounded Sequences

Definition of Monotonic Sequence

A sequence $\{a_n\}$ is **monotonic** when its terms are nondecreasing

$$a_1 \le a_2 \le a_3 \le \cdots \le a_n \le \cdots$$

or when its terms are nonincreasing

$$a_1 \ge a_2 \ge a_3 \ge \cdots \ge a_n \ge \cdots.$$

EXAMPLE

Determine whether each sequence having the given nth term is monotonic.

$$a_n = 3 + (-1)^n$$

SOLUTION This sequence alternates between 2 and 4. So, it is not monotonic.

EXAMPLE

Determine whether each sequence having the given nth term is monotonic.

$$b_n = \frac{2n}{1 + n}$$

SOLUTION We compare the terms b_n and b_{n+1}

$$b_n = \frac{2n}{1 + n} \overset{?}{<} \frac{2(n + 1)}{1 + (n + 1)} = b_{n+1}$$

$$2n(2 + n) \overset{?}{<} (1 + n)(2n + 2)$$

$$4n + 2n^2 \overset{?}{<} 2 + 4n + 2n^2$$

$$0 < 2$$

Starting with the final inequality, which is valid, you can reverse the steps to conclude that the original inequality is also valid. Hence the sequence $\{b_n\}$ is monotonic (nondecreasing).

EXAMPLE

Determine whether each sequence having the given nth term is monotonic.

$$b_n = \frac{2n}{1+n}$$

SOLUTION There is another way to determine whether the sequence is monotonic or not by considering the function

$$f(x) = \frac{2x}{1+x}, \qquad x \geq 1.$$

We differentiate the function

$$f'(x) = \frac{2}{(1+x)^2}$$

Since the derivative is positive for all $x \geq 1$, the function is increasing and hence the sequence $\{b_n\}$ is monotonic (nondecreasing).

Monotonic and Bounded Sequences

Definition of Bounded Sequence

1. A sequence $\{a_n\}$ is **bounded above** when there is a real number M such that $a_n \leq M$ for all n. The number M is called an **upper bound** of the sequence.
2. A sequence $\{a_n\}$ is **bounded below** when there is a real number N such that $N \leq a_n$ for all n. The number N is called a **lower bound** of the sequence.
3. A sequence $\{a_n\}$ is **bounded** when it is bounded above and bounded below.

EXAMPLE

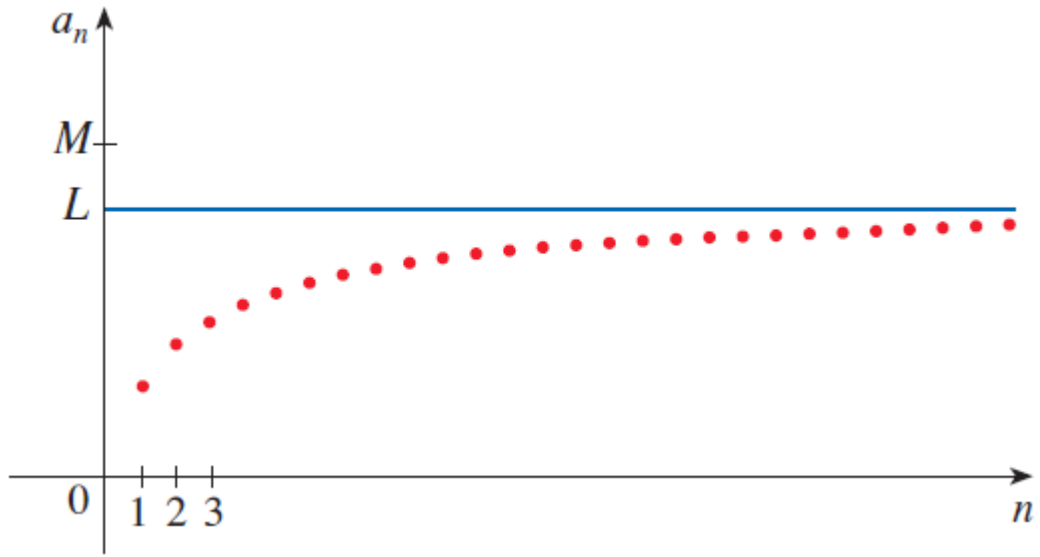
(a) The sequence $\{n\}$ is bounded below but not bounded above.

(b) The sequence $\left\{\frac{n}{n^2+1}\right\}$ is bounded.

Monotonic and Bounded Sequences

12 Monotonic Sequence Theorem Every bounded, monotonic sequence is convergent.

In particular, a sequence that is increasing and bounded above converges, and a sequence that is decreasing and bounded below converges.



Solve exercises of section 9.1.

Start with the suggested problems mentioned in the syllabus.